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OPERATING SYSTEMS

UE24CS242B

Scheduling Algorithms

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- **Slides Credits for all PPTs of this course**
The slides/diagrams in this course are an **adaptation, combination, and enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating System: Three Easy Pieces, Remzi Arpaci



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FCFS and SJF Scheduling

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First - Come, First - Served (FCFS) Scheduling

Process	Burst Time
P1	24
P2	3
P3	3

- Suppose that the processes arrive in the order: P_1 , P_2 , P_3



Waiting time for $P_1 = 0$; $P_2 = 24$; $P_3 = 27$



- Suppose that the processes arrive in the order: P_2 , P_3 , P_1



- Waiting time for $P_1 = 6$; $P_2 = 0$; $P_3 = 3$
- Average waiting time: $(6 + 0 + 3) / 3 = 3$
- The average waiting time under an FCFS policy is generally not minimal, and may even be substantial if the processes' arrival times are not ordered by their execution times.***

- **Convoy effect** - short process behind long process
 - Consider one CPU-bound and many I/O-bound processes
- FCFS scheduling algorithm is non-preemptive. Once the CPU has been allocated to a process, that process keeps the CPU until it releases the CPU
- FCFS algorithm is thus particularly troublesome for time-sharing systems, where it is important that each process to get a share of the CPU at regular intervals.
- It is not desirable to allow one process to keep the CPU

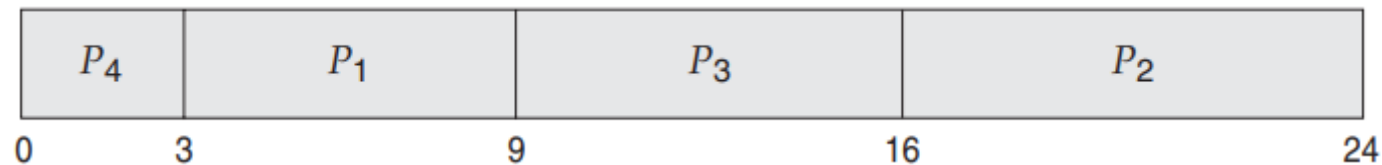


- Associate with each process the length of its next CPU burst
 - Use these lengths to schedule the process with the shortest time
- SJF is optimal – gives minimum average waiting time for a given set of processes
 - The difficulty is knowing the length of the next CPU request

Example of SJF Scheduling

Process	Burst Time
P1	6
P2	8
P3	7
P4	3

- SJF scheduling chart



- Average waiting time = $(3 + 16 + 9 + 0) / 4 = 7$

Note: If FCFS scheduling is used, average waiting time = $(0 + 6 + 14 + 21) / 4 = 10.25$ ms.

- Can be estimated using the length of previous CPU bursts, using exponential averaging

$$\tau_{n+1} = \alpha t_n + (1 - \alpha)\tau_n$$

where τ_{n+1} is the predicted value for the next CPU burst,

The parameter α controls the relative weight of recent and past history in the prediction.

Commonly, $\alpha = 1/2$, so recent history and past history are equally weighted

The value of t_n contains most recent information

The value τ_n stores the past history

Calculate the exponential averaging with $T_1 = 10$, $\alpha = 0.5$ and the algorithm is SJF with previous runs as 8, 7, 4, 16.

Initially $T_1 = 10$ and $\alpha = 0.5$ and the run times given are 8, 7, 4, 16 as it is shortest job first,

So the possible order in which these processes would serve will be 4, 7, 8, 16 since SJF is a non-preemptive technique.

So, using formula: $T_2 = \alpha \cdot t_1 + (1 - \alpha) T_1$

we have,

$$T_2 = 0.5 \cdot 4 + 0.5 \cdot 10 = 7, \text{ here } t_1 = 4 \text{ and } T_1 = 10$$

$$T_3 = 0.5 \cdot 7 + 0.5 \cdot 7 = 7, \text{ here } t_2 = 7 \text{ and } T_2 = 7$$

$$T_4 = 0.5 \cdot 8 + 0.5 \cdot 7 = 7.5, \text{ here } t_3 = 8 \text{ and } T_3 = 7$$

Src: <https://www.geeksforgeeks.org/shortest-job-first-3-cpu-scheduling-with-predicted-burst-time/>

$$T_5 = 0.5 \cdot 16 + 0.5 \cdot 7.5 = 11.25, \text{ here } t_4 = 16 \text{ and } T_4 = 7.5$$

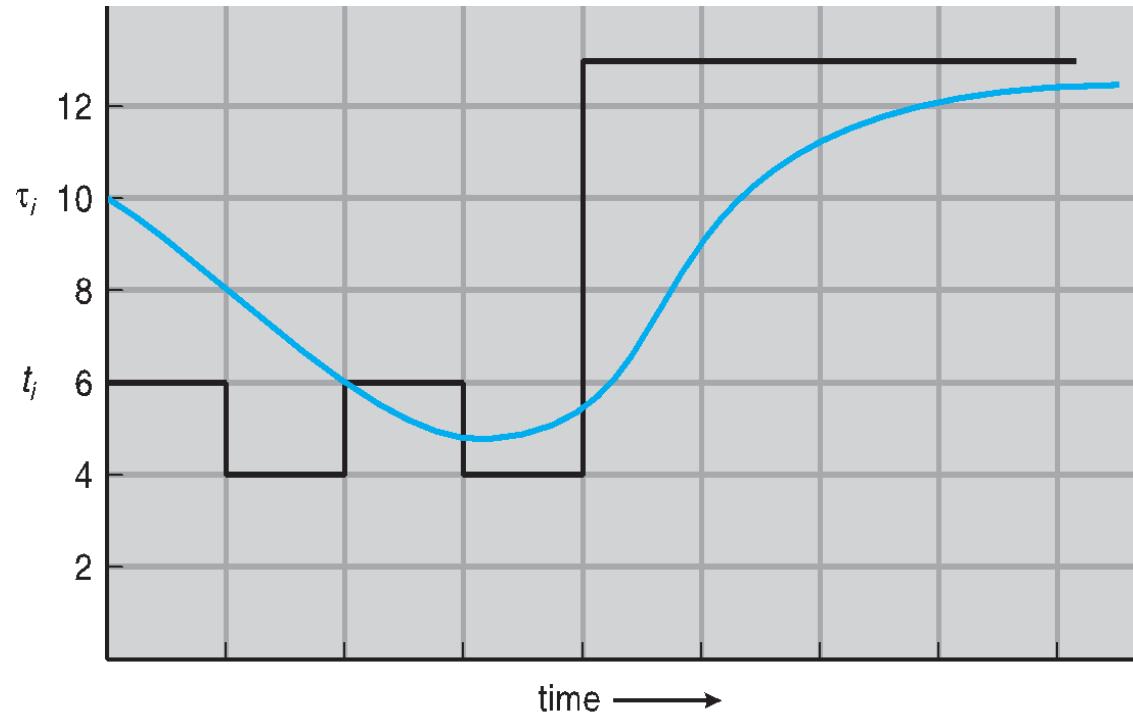
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Prediction of the Length of the Next CPU Burst



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$$\tau_{n+1} = \alpha t_n + (1 - \alpha)\tau_n$$

CPU burst (t_i)	6	4	6	4	13	13	13	...	
"guess" (τ_i)	10	8	6	6	5	9	11	12	...

$$\tau_{n+1} = \alpha t_n + (1 - \alpha)\tau_n$$

- $\alpha = 0$
 - $\tau_{n+1} = \tau_n$
 - Most recent CPU burst does not count
- $\alpha = 1$
 - $\tau_{n+1} = \alpha t_n$
 - Only the actual last CPU burst counts
- If we expand the formula, we get:
 - $\tau_{n+1} = \alpha t_n + (1 - \alpha)\alpha t_{n-1} + \dots$
 - $\quad \quad \quad + (1 - \alpha)^j \alpha t_{n-j} + \dots$
 - $\quad \quad \quad + (1 - \alpha)^{n+1} \tau_0$
- Since both α and $(1 - \alpha)$ are less than or equal to 1, each successive term has less weight than its predecessor

Suppose that the following processes arrive for execution at the times indicated. Each process will run for the amount of time listed. In answering the questions, use non-preemptive scheduling, and base all decisions on the information you have at the time the decision

<u>Process</u>	<u>Burst Time</u>
P_1	8
P_2	4
P_3	1

- What is the average wait time for these processes with the FCFS scheduling algorithm?

$$\text{Average wait time} = (0 + 8 + 12) / 3 = 6.67$$

- What is the average wait time for these processes with the SJF scheduling algorithm?

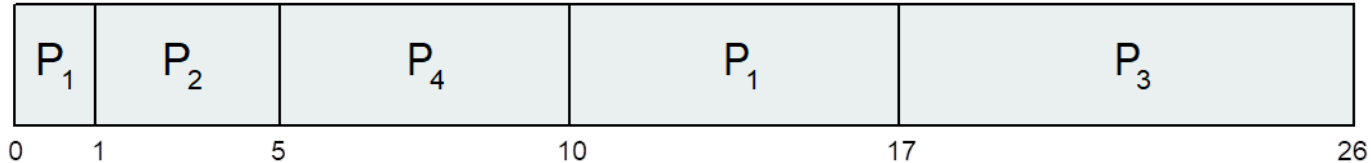
$$\text{Average wait time} = (5 + 1 + 0) / 3 = 2$$

Example of Shortest-remaining-time-first

- Preemptive SJF Scheduling is sometimes called SRTF
- Now we add the concepts of varying arrival times and preemption to the analysis

<u>Process</u> <u>Time</u>	<u>Arrival Time</u>	<u>Burst</u>
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

- Preemptive SJF Gantt Chart



- Average waiting time = $[(10-1) + (1-1) + (17-2) + (5-3)] / 4 = 26/4 = 6.5 \text{ msec}$



THANK YOU

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